Proof Audit: Supervised Denoiser Dynamics

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Assurance

Native same-family review: provisional. No cross-family semantic acceptance, theorem novelty or architecture improvement is claimed. Actual returned model identity was unavailable; the requested reviewer route is recorded separately.

Scope

All nine source units U0-U8 and 35 extracted microclaims were checked, then rechecked in the same session. Source-unit execution coverage is complete; coverage does not prove that all possible obligations were discovered.

Before and after

R1: an implicit conditional-mean version was replaced by an explicit smooth Bayes-ratio representative. R2: density and vector-domain notation were clarified. The U6-U4 dependency was recorded. No mathematical formula, prior class or numerical tolerance changed.

Reviewed mathematical relations

The supervised conditional mean obeys a necessary heat/denoiser PDE. It need not be invariant along the marginal flow. A wrong-prior denoiser can satisfy the PDE. A precision derivative generally includes a contracted third posterior moment. These are bounded mathematical statements, not a new model result.

Separate numerical verification

All 120 fixed existing-law probes and 240 central-difference checks pass their registered gates. Complete derivative arrays and both step sizes remain available. Numerical reproduction and prior-file application were outside the semantic reviewer scope.

Remaining scope

No new architecture, PDE regularizer, target change or covariance rescue is selected. Report1 and all previous data, training checkpoints and P histories are unchanged. The useful-ODE-successor objective remains open.

Build provenance

This PDF is rendered with Matplotlib. A separate LaTeX summary source is supplied, but no local TeX compiler was found and no TeX compilation is claimed.